

PINN–PINO TwinMoE: Physics-Constrained Expert Coordination for Multiscale Battery Digital Twins

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Code and training: <https://github.com/fatasia/DeepMonkey-Studio/tree/main/research/battery-twin>

Abstract

Battery digital twins combine degradation over cycles with electrothermal changes within operating windows, requiring different physical constraints and computational budgets. We present the component architecture of PINN–PINO TwinMoE: a degradation model extended by a single-particle physics head, transient-state estimation with lightweight and spectral-operator experts, and service policies that distinguish executed experts from accepted predictions. We detail physical-loss normalization, attenuation of conflicting data and physics gradients, and conditional execution informed by physical risk. In a recorded 77-cell HUST analysis, mean normalized RMSE decreases from 0.166026 to 0.162743 and three trajectory-shape metrics improve, while mean MAPE increases from 14.165708% to 14.753008%. A separate 63-scenario synthetic test records voltage MAE of 0.018221 V. These component-level results motivate evaluating physical constraints through error, trajectory shape, and operating cost together. Public source provides physics-head and operator training entry points. A health-context interface connects the temporal tasks, but its joint coupling benefit has not yet been evaluated.

1 Introduction

A battery twin that always invokes its most expensive model may waste computation; one that always selects its fastest model may miss a change in operating conditions or physical mismatch. Long-horizon degradation and transient-state estimation also require distinct targets. We study how experts can specialize while separating model computation, physical constraints, and the decision to accept a result.

The architecture has three levels. The degradation model consumes historical cycles and cell conditions, with a physics head imposing concentration-field and effective-parameter constraints. The transient model combines a temporal representation, radial spectral operators, and a lightweight expert. Service policies assess domain validity and disagreement before selecting an accepted prediction. Health context is an explicit interface between temporal tasks; current results evaluate these tasks separately.

The contribution is this component architecture, inspectable physical-training and conditional-execution mechanisms, and an analysis of the tradeoffs in recorded experiments. The concrete choices include normalized physical losses, gradient-conflict attenuation on shared parameters, lightweight/operator mixing, and execution conditions informed by physical risk. We distinguish implementation of these mechanisms from measured component outcomes and from untested joint benefits.

2 Related Work and Design Position

BatteryMFormer supplies the degradation-trajectory backbone [3]. PINO already combines data with differential-equation constraints [1], and PIMOE studies physics-informed experts for battery degradation [2]. Our design focuses on physical extensions, computation division between transient experts, and deployment-time result adoption. DeepSeekMoE makes expert specialization concrete through architectural comparisons [4]; our lightweight and spectral experts instead have different computational structures for a physical task. Gradient interference also has established treatments, including PCGrad’s gradient projection [5]; the implementation here attenuates a physical-loss weight.

3 Implemented Method

3.1 Task Interfaces and Expert Roles

Let H denote historical cycle observations, X operating-window inputs, and h health information available before that window. The long-horizon expert outputs degradation quantities and effective physical parameters; the transient expert estimates concentration, voltage, and temperature within a window. Their targets and training data differ. The dashed interface in Figure 1 denotes health conditioning, not a demonstrated jointly trained model.

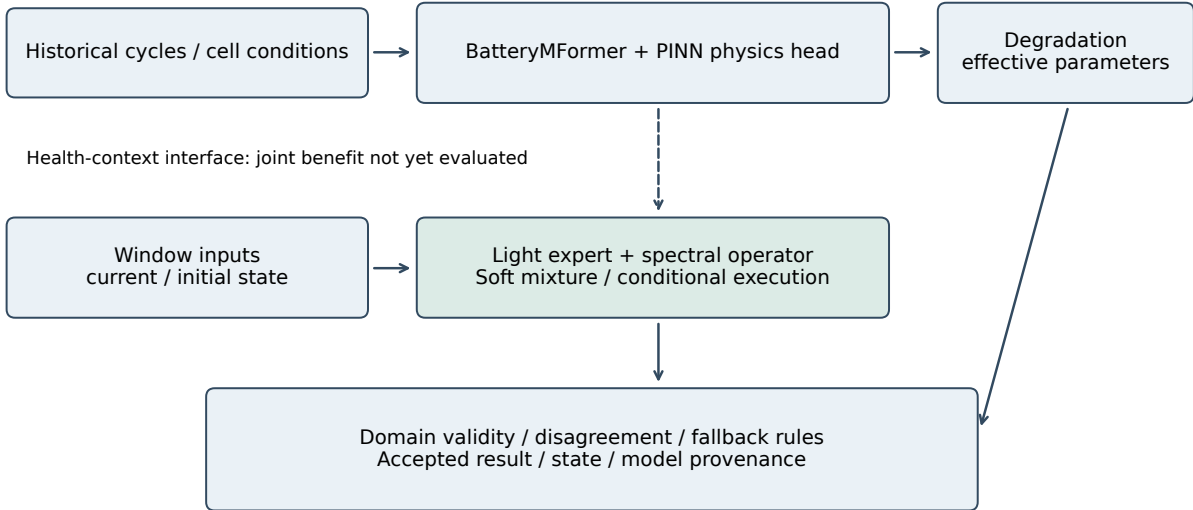


Figure 1: Model computation is separated from result adoption. The dashed health-context interface exists, but its cross-task benefit has not been evaluated.

3.2 Long-Horizon Physics-Informed Expert

The implementation augments the BatteryMFormer backbone with **BatterySPMPINNHead**. Cell and measured-curve context inform concentration and effective physical parameters. Its single-particle

constraints address radial diffusion, surface flux, conservation, initial conditions, and voltage consistency. A schematic radial diffusion residual is

$$\mathcal{R}_c = \partial_t c - \frac{D_{\text{eff}}}{r^2} \partial_r (r^2 \partial_r c),$$

with coordinate scaling and effective parameters defined by the implementation. Center/surface boundary conditions and integral constraints complement interior collocation. Additional terms include lithium inventory, rest-weighted open-circuit-voltage anchoring, chemistry priors, aging dynamics, and observed physical quantities.

The training objective combines prediction and physical terms. Each physical loss is scaled by its detached current absolute magnitude with a lower clamp. Diagnostics measure the cosine between data and physics gradients on shared parameters; conflicting physics pressure is attenuated toward a configurable minimum. This is loss-weight attenuation, not gradient projection. Effective normalized diffusion parameters require independent geometry and electrochemical information before interpretation as intrinsic material diffusivities.

Physical losses and gradient pressure. For physical terms ℓ_k , the implemented normalization is

$$L_P = \sum_k a_k \frac{\ell_k}{\text{stopgrad}(\text{clip}(|\ell_k|, 10^{-6}, 10^6))}. \quad (1)$$

Here a_k are configured weights. Stopping gradients through the denominator scales the gradient contribution of each physical term. On parameters reached by both data and physics losses, let their gradients be g_D, g_P . The diagnostic computes

$$\rho = \frac{\langle g_D, g_P \rangle}{\|g_D\| \|g_P\|}, \quad \alpha = 1 - (1 - \alpha_{\min}) \max(0, -\rho). \quad (2)$$

Missing or negligible overlapping gradients give $\alpha = 1$. The diagnostic defaults to a minimum of 0.15; enabled training paths can configure the update interval and smoothing. These equations specify an implemented mechanism, not identical settings across all archived checkpoints. It retains physical pressure for compatible gradients and reduces opposition; residual reduction and effective-parameter identifiability still require separate evaluation.

3.3 Short-Horizon Physics-Informed Operator

SPMPINOTransformer combines a temporal representation with radial spectral blocks and electrothermal outputs. Let $f_L(x)$ and $f_O(x)$ denote lightweight and operator concentration predictions and $w(x) \in [0, 1]$ the learned operator weight. The soft mixture is

$$\hat{c}(x) = (1 - w(x))f_L(x) + w(x)f_O(x).$$

In **dynamic** mode both experts execute. In **dynamic_sparse**, the lightweight expert executes first and the operator runs when physical risk or learned probability exceeds its configured threshold. The native ONNX export currently uses **dynamic**.

The model accepts optional health/degradation context. Inspected training paths construct this context from capacity ratio and current/voltage statistics. The architecture therefore offers a conditioning interface across time scales; a measured handoff from the long-horizon PINN output to the short-horizon PINO remains an experiment to complete. The framework does not assume joint end-to-end training of all experts.

3.4 Three Levels of Expert Coordination

Level	Decision and retained information
Intra-model	Learned light/operator weights; sparse mode also uses physical-risk thresholds.
Life prediction	Rules inspect extrapolation, confidence, domain warnings, and standard/PINN disagreement before selecting the accepted life result.
Twin runtime	Rules select qualified PINO, electrothermal baseline, or physical SPM fallback, retaining alternative predictions and model identity.

The life policy flags high extrapolation, weak fit, or inadequate confidence. Large disagreement requires both an absolute and a relative threshold. The twin runtime extracts bounded terminal SOC and temperature for committed state. Logs distinguish executed experts from the adopted expert, which supports analysis of rejected predictions and fallback costs.

Execution versus adoption. Let $r(X)$ be normalized physical risk and $p_O(X)$ the learned operator-execution probability. The sparse call condition can be written as

$$z(X) = \mathbf{1}\{r(X) > \tau_r \vee p_O(X) > \tau_p\}.$$

Thresholds and features are configuration dependent. With $z = 0$, operator computation is skipped; $z = 1$ requests computation but does not require the service to adopt its output. Service policies subsequently check model version, domain, confidence, and disagreement while retaining alternatives. This separates computational budgeting from runtime result selection. Call fraction, actual latency, error, and rejection rate must therefore be measured separately.

3.5 Deployment

The platform has separate native ONNX model packages for life and transient inference. A Rust host handles the deployed inference path without a Python/PyTorch runtime dependency. Runtime provenance and domain checks accompany predictions. Numerical agreement between exported and research models, request latency, and handoff behavior are separate evaluation targets from predictive accuracy.

4 Data and Evaluation Protocol

The inspected frozen synthetic comparison records 1,255 training, 127 validation, and 63 test scenarios from an LFP electrothermal dataset. These are simulation scenarios, not 1,445 independently measured physical cells. Group assignments, generation parameters, failures, and source-cell identifiers require verification from the dataset manifest before final publication.

The inference task must also be explicit. The comparison script calls an evaluation function whose model wrapper defaults to **observer**. The wrapper’s separate **forecast** mode restricts voltage context to a historical prefix. Thus, the existing observer-path results cannot, by themselves, establish causal future prediction. Whether the selected checkpoint actually consumes the available voltage context must be inspected from its configuration. Final experiments will separately evaluate observer inference and historical-prefix forecasting with a declared prediction cutoff and known-control assumptions.

4.1 Routing Evaluation Semantics

The archived evaluator derives reference labels from expert outputs returned by the current execution mode. In sparse mode, unexecuted operator outputs are replaced by lightweight outputs. The reference consequently changes across policies, so we exclude those routing-accuracy scores from policy comparisons. A valid comparison must evaluate both frozen experts offline for every sample and use one common reference before separately measuring online error and cost:

$$y^*(X) = \operatorname{argmin}_{e \in \{L, O\}} [\ell_e(X) + \lambda C_e].$$

This equation specifies an evaluation requirement, not a newly completed experiment. The voltage, concentration, temperature, and request-time results below retain their original measurement scope. Request and batch latency are reported separately; call fraction is not converted directly into speedup.

5 Recorded Evidence

These values are transcribed from existing artifacts; training was not rerun for this draft. The July 2026 frozen comparison uses checkpoint `spm-pino-transformer-lfp-v10-6-speed-first`. It must not be silently combined with later guarded or transfer checkpoints.

Recorded metric	Dynamic	Dynamic sparse
Concentration MAE (source normalization)	0.008161	0.008711
Voltage MAE (V)	0.018221	0.018236
Temperature MAE (°C)	0.159237	0.159237
Mean request latency (ms)	5.955382	5.675562
Request P95 latency (ms)	8.588700	7.843700

Routing-accuracy values are omitted because the archived labels change with execution mode. Request timing contains 189 repeated calls rather than 189 independent scenarios. Recorded batch P95 latency is 7.007 ms for dynamic and 32.240 ms for sparse, showing that request and batch measurements cannot support a blanket speedup claim. The timing environment and protocol need revalidation.

The three-optimization-seed summary records voltage MAE mean 0.017967 V and sample standard deviation 0.001933 V; concentration MAE has coefficient of variation 17.82%. The seed-2026 checkpoint is named speed-first, whereas the additional checkpoints are named full-seed variants. Configuration identity must be checked before treating this table as a strictly controlled seed-only experiment. These rows also do not establish three-seed common-oracle sparse-routing accuracy.

5.1 Degradation and Physical Consistency

The recorded HUST analysis uses 77 cell IDs, 20,000 bootstrap resamples, and six Holm-corrected paired tests. Table 1 reports cell-aggregated results in the source normalization. Its source-level holdout designation still requires a final audit of upstream training and calibration membership. The mean RMSE difference is -0.003283 , with bootstrap 95% interval $[-0.005754, -0.000627]$ and Holm-adjusted paired-test $p = 0.00644$. Each of the 77 cells favors the PINN on the three recorded shape metrics. Mean MAPE instead increases by 0.587300 percentage points. The results show a tradeoff between pointwise relative error and trajectory shape.

Metric	Standard	PINN
RMSE (normalized)	0.166026	0.162743
MAPE (%)	14.165708	14.753008
Monotonicity violation	0.364614	0.337601
Excess RMS	0.006793	0.005608
Smoothness RMS	0.018637	0.015769

Table 1: Recorded HUST cell-level analysis; lower is preferable for these metrics.

Three internal PINN optimization runs record MAPE values of 1.350986%, 1.3208274%, and 1.3223664%; their mean is 1.3313933%, versus the recorded standard model’s 1.2129676%. Removing chemistry conditioning worsens MAPE in all three paired runs, by an average 0.0584 percentage points. The parameter-freezing ablation reports large surface-flux errors, illustrating why prediction error alone is insufficient for evaluating physical identification.

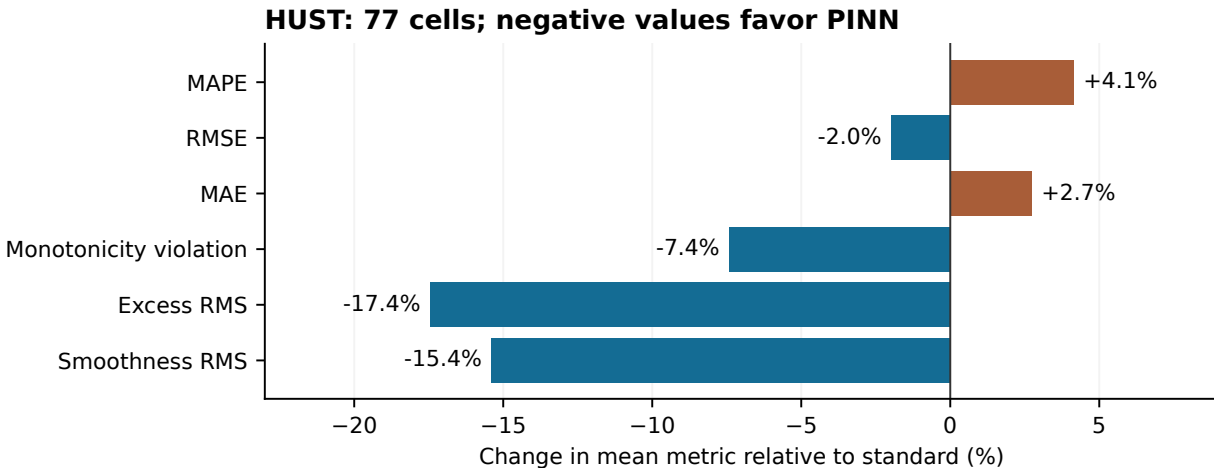


Figure 2: Relative changes computed from the recorded means in the 77-cell HUST analysis. Bars are ratios of aggregate means, not mean per-cell percentage changes; no confidence intervals are implied.

6 Limitations and Remaining Experiments

Synthetic concentration labels do not validate internal concentration fields in measured cells. Near-zero SOC error may reflect shared conservation-based construction rather than sensor-grounded accuracy. Local Jacobian perturbations do not prove global stability. Two large-capacity target cells do not establish broad cross-domain transfer. A one-pack graph residual result that does not beat a simple baseline does not establish pack generalization.

The remaining joint study will compare always-standard, always-physics, rule-only, learned-only, and combined policies on cell-isolated tasks. It will freeze counterfactual expert labels, declare forecast cutoffs, and ablate health context. Paired error and shape metrics will be reported alongside operator calls, rejection rates, state continuity, and synchronized request/batch latency. Measured-voltage targets and source membership must be frozen before final evaluation.

7 Reproducibility and Disclosure

Training source is available in [DeepMonkey-Studio/research/battery-twin](https://github.com/DeepMonkey-Studio/research/battery-twin). The directory contains the PINN physics head, PINO model, training scripts, generated scenarios, tests, and per-file source hashes. Its README provides installation and small training commands. Platform expert adoption is implemented in [batteryExpertRouting.ts](#). Full BatteryMFormer fine-tuning additionally requires upstream adaptations, data, and weights; the published source snapshot is distinct from the archived experimental checkpoints.

Existing evidence is indexed by source path and SHA-256. The reviewed seven-item experiment manifest matches current artifact bytes. Source revision: 501e430095b5 (full hash in the evidence index). Existing local changes make per-file hashes necessary. A public release needs finalized code/data permissions, frozen dependencies, checkpoint hashes, executable evaluation commands, and regenerated result tables.

Generative AI assisted source inspection, organization, and manuscript drafting.

8 Conclusion

PINN–PINO TwinMoE separates degradation constraints, lightweight/operator transient computation, and service-level adoption. The public implementation makes loss normalization, gradient-conflict attenuation, and conditional execution inspectable. Recorded data show that improvements in trajectory metrics can coexist with worse relative error, and fewer expert calls do not imply batch speedup. The report consequently separates accuracy, physical shape, computation, and result adoption, leaving the joint benefit of health conditioning to controlled evaluation on common tasks.

References

- [1] Z. Li et al. Physics-Informed Neural Operator for Learning Partial Differential Equations. arXiv:2111.03794. <https://arxiv.org/abs/2111.03794>.
- [2] X. Huang et al. Physics-informed mixture of experts network for interpretable battery degradation trajectory computation amid second-life complexities. arXiv:2506.17755, 2025. <https://arxiv.org/abs/2506.17755>.
- [3] R. Tan et al. BatteryMFormer: Multi-level Learning for Battery Degradation Trajectory Forecasting. arXiv:2605.27044, 2026. <https://arxiv.org/abs/2605.27044>.
- [4] D. Dai et al. DeepSeekMoE: Towards Ultimate Expert Specialization in Mixture-of-Experts Language Models. arXiv:2401.06066, 2024. <https://arxiv.org/abs/2401.06066>.
- [5] T. Yu et al. Gradient Surgery for Multi-Task Learning. NeurIPS, 2020. <https://arxiv.org/abs/2001.06782>.